

ARTICLE

# SMART REAL-TIME STRESS MONITORING SYSTEM USING AI AND IoT

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**ABSTRACT:** Chronic stress is prevalent in today's hectic society, and many wearables and accessories have made it easier to detect and monitor it effectively. Many erratic bodily functions and physiological factors can help accurately deduce stress with the help of 'Relaxo-Monitor'. Recent works have demonstrated that combining behavioral and physiological data can detect stress with higher accuracy. This paper explains how the proposed system uses Internet of Things (IoT) sensors for (Beats per Minutes) BPM, temperature, voice stress markers, movement, and facial tracking to accurately diagnose stress. The monitor employs Firebase for user authentication and cloud storage of raw data; a Flask-based backend is connected to the UI, which is built using HTML, CSS, and JavaScript. The monitoring system provides an interactive analytics dashboard for visualizing vital statistics and can generate personalized reports as exportable summaries and also send the report through Short Message Services (SMS) using Twilio. An AI component analyzes the accumulated results to suggest preventive actions (e.g., breathing exercises) when high stress is detected. The system combines multiple indicators to improve accuracy. Experimental trials and multiple evaluations have shown that the proposed system can effectively sort individuals into different stress categories based on combined sensor data. The final product is an easy-to-use, affordable solution for academic settings, corporate environments, and basic healthcare monitoring.

**KEYWORDS:** Stress; heart rate; machine learning, Stress Monitoring, Real-Time Stress Detection, Internet of Things (IoT);

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## 1 Introduction

Psychological stress is a leading concern in healthcare and wellness. Stress can be induced and manifests without overt external indicators. Individuals may project a composed and calm behavior while experiencing severe internal cognitive and emotional fatigue. In recent years, chronic stress from heavy workloads, social and academic pressure can manifest in subtle physiological and behavioral changes (e.g., increased heart rate, sweating) that go unnoticed until they cause fatigue and anxiety, which is detrimental to health in the long term. Many individuals ignore these symptoms until serious physical or psychological conditions require medical intervention.

Conventional assessments, such as surveys and questionnaires, lack real-time responsiveness and continuous monitoring capabilities, highlighting the need for continuous monitoring devices. Wearable devices, including smartwatches and chest straps, routinely capture heart rate, skin conductivity, and

motion data; these physiological signals correlate with autonomic stress responses [1][3]. The integration of Internet of Things (IoT) and Machine Learning (ML) technologies within the healthcare sector provides significant opportunities for continuous physiological signal monitoring, personalized interventions, and real-time feedback mechanisms.

This paper presents Relaxo-Monitor, a real-time stress detection system that integrates multiple modalities. The system continuously acquires physiological (heart rate, temperature), behavioral (facial emotion, voice tone, movement), and these features are analyzed using ML-based analytical methods to identify stress patterns. The system incorporates Firebase authentication, Flask based backend architecture, a live analytics dashboard, and automated report generation and SMS functionalities using Twilio. The primary objective of this study is to develop an accessible and efficient stress monitoring framework capable of supporting continuous real-time stress assessment through AI/ML and IoT technologies.

## 2 Related works

Recent IoT-based stress-monitoring systems leverage wearable sensors (e.g. heart rate/HRV, galvanic skin response, accelerometer) and ML to detect stress in real time. Typical architectures preprocess signals to extract features (time/frequency HRV metrics, EDA peaks, MFCC voice features, facial embeddings), then apply classifiers (RF, SVM, CNNs, etc.) on-device or in the cloud. Key trade-offs involve sensor modality (battery vs information), model complexity (latency vs accuracy), and deployment (edge inference for privacy vs cloud for heavy analytics). We compare at least eight recent works (2018–2026) in the table below, summarizing their sensors, ML methods, deployment, evaluation, strengths, and weaknesses.

The work by Pataca *et al.* (2025) is a systematic review emphasizing that electrodermal activity (EDA), photoplethysmography (PPG/HRV), and accelerometer data dominate wearable stress studies. Commonly extracted features include HRV time- and frequency-domain metrics and skin-conductance levels. This review notes that Random Forests, SVMs, and deep neural nets have achieved high accuracy (up to ~99%) in controlled settings, but it highlights a lack of large, standardized datasets and limited real-world validation.

Fortunato *et al.* (2026) developed a *multimodal wearable device* integrating ECG, GSR, skin temperature (ST), and sweat-cortisol sensing. Their device is designed for continuous stress assessment and subtyping. The platform collects raw ECG and GSR signals via flexible skin electrodes; ST and biochemical (cortisol) sensors provide additional context. Although no ML details are provided, the contribution is in hardware integration – addressing specificity issues by combining physiological and molecular biomarkers. A limitation is that it is a prototype (not an evaluated ML model) and requires specialized hardware and calibration.

Kumar *et al.* (2026) collected chest-ECG, wrist-EDA (GSR), and skin temperature data from adult women during controlled stressors. They filtered ECG and extracted 11 HRV features, decomposed GSR via wavelets (11 features), and derived 11 ST features using a Butterworth filter. These 33 features were standardized and fed into a deep neural network model (details unspecified). They split data 75/25 for train/test. Report included high classification performance for “no stress vs acute vs chronic stress,” though exact accuracies were unspecified in the abstract. Strengths: multimodal signals and thorough preprocessing; limitations: small controlled study (N=?) and lack of real-time deployment.

Kankal *et al.* (Sensors 2023) presented an IoT stress-monitoring platform called *Stress-Track* that used body-humidity, body temperature, and step count from wearables. A cloud-based ensemble ML

model (Random Forest with 100 trees, and gradient boosting) was trained on these features. They report ~80% accuracy for stress level classification on a curated dataset (labeled “Stress-Lysis” with 2001 samples). Deployment is offline/cloud: wearables send data to the cloud server for analysis. Strength: uses unobtrusive environmental sensors; limitation: these are indirect stress proxies (no direct HR or EDA), and no on-device inference, so real-time alerts depend on connectivity.

Campanella *et al.* (Sensors 2023) evaluated stress classification from an Empatica E4 wristband’s PPG (blood volume pulse) and EDA. They extracted 27 features from 29 subjects’ data and compared Random Forest, SVM, and logistic regression classifiers. The RF achieved the best results (~76.5% accuracy; F1 ~0.65). All processing was done offline in WEKA/MATLAB (no real-time deployment). Key points: uses two physiological sensors, standard feature ranking (chi-sq, correlation), and reports modest accuracy. Limitation: small dataset, binary stress labelling, and relatively low recall. This work exemplifies classic ML on multimodal signals with feature engineering.

Khayyat *et al.* (2024) proposed a real-time stress detector using just wrist HR (HRV). They designed a Capsule Network (CapsNet) to classify stress levels from HRV features. The model was intended to run in real time on-device. They reported improved detection over CNNs (exact metrics unspecified). Strength: novel DL architecture for HRV; limitation: only heart-rate is used (no EDA), and exact performance numbers and dataset size were unspecified. This reflects a trend of using a single affordable sensor with a powerful model to simplify deployment, at the expense of ignoring other modalities.

Sharma *et al.* (Ta, 2025) developed *mHELP*, a mobile stress app for students. Each participant wore an Apple Watch streaming heart rate and accelerometer data to an iPhone app. A proprietary on-device ML algorithm flagged stress episodes in real time, triggering relaxation prompts. Although details are sparse, this system illustrates an edge-deployed approach: a lightweight stress model (type unspecified) running on a paired watch+phone, with no cloud dependence mentioned. They conducted a trial showing reduced self-reported stress. Strength: real-time feedback and integrated UX; limitation: validation was on college subjects only, and the ML model specifics (features, accuracy) are not disclosed.

Indani & Vyawahare (2026) describe a “Stress Detection System” integrating HR (ECG/PPG), GSR, body temperature, and optionally respiration. Their microcontroller-based platform collects these in real time and classifies stress (low/moderate/high) via supervised ML (SVM, RF, neural nets). They emphasize IoT connectivity: data streams to a mobile/web dashboard for visualization. In a case study, their system reportedly achieved high accuracy (unspecified) in a university setting. Strengths: multi-sensor real-time design and ML analysis; limitations: being a classroom project, precise model performance and dataset size are not rigorously presented, and sensors like GSR require calibration.

### 3 Methods Used

Real-time monitoring: RelaxoMonitor processes sensor data in real-time. Using sensors and input devices, the system streams data such as heartbeats (BPM and HRV) via MAX30102 Pulse Oximeter Heart Rate Sensor Module. BPM is calculated using the inverse relationship between the RR interval and the number of beats per minute.

$$BPM = \frac{60000}{RRms} \text{ or } RRms = \frac{60000}{BPM} \quad (1)$$

Where RRms represents the interval between two consecutive heartbeats measured in milliseconds

Heart rate variability (HRV), which represents the variation between consecutive RR intervals, is calculated as,

$$HRV = |RR_{current} - RR_{previous}| \quad (2)$$

A lower HRV may indicate stress or fatigue, while higher HRV is generally associated with a relaxed physiological state.

Temperature monitoring is performed using the DHT11 sensor. The collected temperature values are compared against predefined threshold conditions to identify abnormal physiological responses.

$$Stress\_temp = \{HIGH, T > 37.5^{\circ} C; NORMAL, T \leq 37.5^{\circ} C\} \quad (3)$$

facial expression frames are analyzed via a laptop-integrated camera and CNN based DeepFace emotion recognition library. The system captures facial frames and classifies emotions such as happy, neutral, sad, angry or fearful. Emotional expressions associated with negative expressions are considered as stronger indicators of stress

$$EmotionScore = f(angry, sad, fearful, neutral, happy) \quad (4)$$

Movement data is collected using accelerometer sensor readings. Consecutive frames are compared to estimate the level of physical movement and restlessness

$$MovementScore = mean(|Frame_t - Frame_{t-1}|) \quad (5)$$

Where Frame\_t represents the current frame and Frame\_(t-1) represents the previous frame.

Microphone input is used for voice stress. The audio is continuously sampled and processed to calculate the Root Mean Square (RMS) amplitude, which provides an estimate for vocal intensity.

$$Volume = \sqrt{(1/N) * \sum(x[n]^2)} \quad (6)$$

e.g., A sudden heart rate elevation combined with an increase in body temperature and an anxious facial expression may detect possible stress.

In RelaxoMonitor, streaming data is buffered in short windows, features such as BPM, emotion, temperature, etc. are extracted, and a trained model using logistic regression outputs stress likelihood at regular intervals.

$$StressLevel = f(BPM, HRV, Temp, Emotion, Voice, Movement) \quad (7)$$

The dashboard then logs a stress event when a threshold is exceeded, and an alert is sent (e.g., prompting deep-breathing exercises).

Firestore authentication: Users can access their data through Firestore Authentication. Each user logs in with an email/password or third-party credentials (Google, Apple) managed by Firestore. On login, a unique user ID is created, and all subsequent data (sensor streams, stress logs, reports) are linked to this ID in the cloud. This ensures that all user data is securely personalized and partitioned. Firestore's authentication also provides OAuth flows and token-based sessions, so the mobile app can seamlessly

communicate with Firebase Cloud Firestore while the user is signed in. Firebase has built-in industry-standard security (HTTPS communication, password hashing) without building a custom auth system.

**Analytics dashboard:** RelaxoMonitor also includes an interactive analytics dashboard accessible via the web. The dashboard helps in visualizing the collected data in real time and over historical periods. Charting libraries such as Chart.js are used to plot time series of vital signs (heart rate, temperature), as well as stress-level indicators. e.g., a heart rate curve and a stress score timeline are displayed, so users can see different physical signals with stress events. The dashboard queries data from the cloud datastore (Firebase) to update charts. The dashboard also computes summary statistics (e.g. average heart rate per day) and shows them as gauges or histograms.

**Report generation:** RelaxoMonitor can compile user data into a stress report through a button. The function queries the cloud database for the user’s recent stress log and summary statistics. It then formats a PDF (or HTML) report including the user’s name, date range, key metrics such as max BPM, number of stress events, average temperature, etc. The report highlights any trends (e.g. rising stress over days) and suggests corrective actions (e.g. “Your heart rate tends to spike around lunchtime – consider taking a short walk.”). Generated reports are saved to cloud storage under the user’s account and can be emailed or downloaded.

**AI suggestions/recommendations:** Beyond detecting stress, RelaxoMonitor provides recommendation-based feedbacks to users. When a stress event is detected, the system use an ML-driven suggestion engine using OpenAI. For instance, high stress with elevated heart rate may prompt a guided breathing exercise; high stress with significant movement might suggest resting. In RelaxoMonitor, after each stress detection, a brief text delivered on the screen (e.g. “Your breathing has sped up. Try this 2-minute breathing guide.”).

## 4 Result and Discussion

RelaxoMonitor was developed for evaluating real-time stress monitoring using multiple physiological and behavioral inputs. The system successfully integrated heart rate monitoring, temperature sensing, facial emotion analysis, motion tracking, and voice-based stress analysis, displayed on a single monitoring dashboard. Testing showed that the system could process sensor data in real time with minimal delay.

Table 1 presents the classification report obtained from the Logistic Regression-based stress classification model. The model achieved an overall accuracy of 82.76% during testing. Table 2 shows the confusion matrix representing the predicted and actual stress classes.

**Table 1 Classification report**

| Class     | Precision | Recall | F1-Score | Support |
|-----------|-----------|--------|----------|---------|
| 0         | 0.82      | 1.00   | 0.90     | 75      |
| 1         | 0.84      | 0.99   | 0.91     | 70      |
| 2         | 0.00      | 0.00   | 0.00     | 29      |
| Accuracy  | -         | -      | 0.83     | 174     |
| Macro Avg | 0.55      | 0.66   | 0.60     | 174     |

|              |      |      |      |     |
|--------------|------|------|------|-----|
| Weighted Avg | 0.69 | 0.83 | 0.75 | 174 |
|--------------|------|------|------|-----|

**Table 2 Confusion matrix**

| Actual \ Predicted | 0  | 1  | 2 |
|--------------------|----|----|---|
| 0                  | 75 | 0  | 0 |
| 1                  | 1  | 69 | 0 |
| 2                  | 16 | 13 | 0 |

**Table 3 Overall model performance**

| Metric            | Value               |
|-------------------|---------------------|
| Accuracy          | 82.76%              |
| Total Samples     | 174                 |
| Model Used        | Logistic Regression |
| Number of classes | 3                   |

During testing, the MAX30102 Pulse Oximeter sensor monitored heart rate while the DHT11 sensor recorded changes in body temperature during a stress-related event. Facial emotion analysis using the integrated camera detected emotional states such as anxiety, sadness, and neutral expressions. Movement data collected through accelerometer-based monitoring helped identify periods of restlessness and irregular motion patterns, which were commonly observed during stress-related events.

The dashboard displayed live sensor readings, stress status indicators, and historical monitoring trends. The system generated automated stress alerts whenever physiological and behavioral parameters exceeded the preset threshold conditions. In multiple test scenarios, elevated heart rate combined with increased temperature and anxious facial expressions resulted in successful stress detection events.

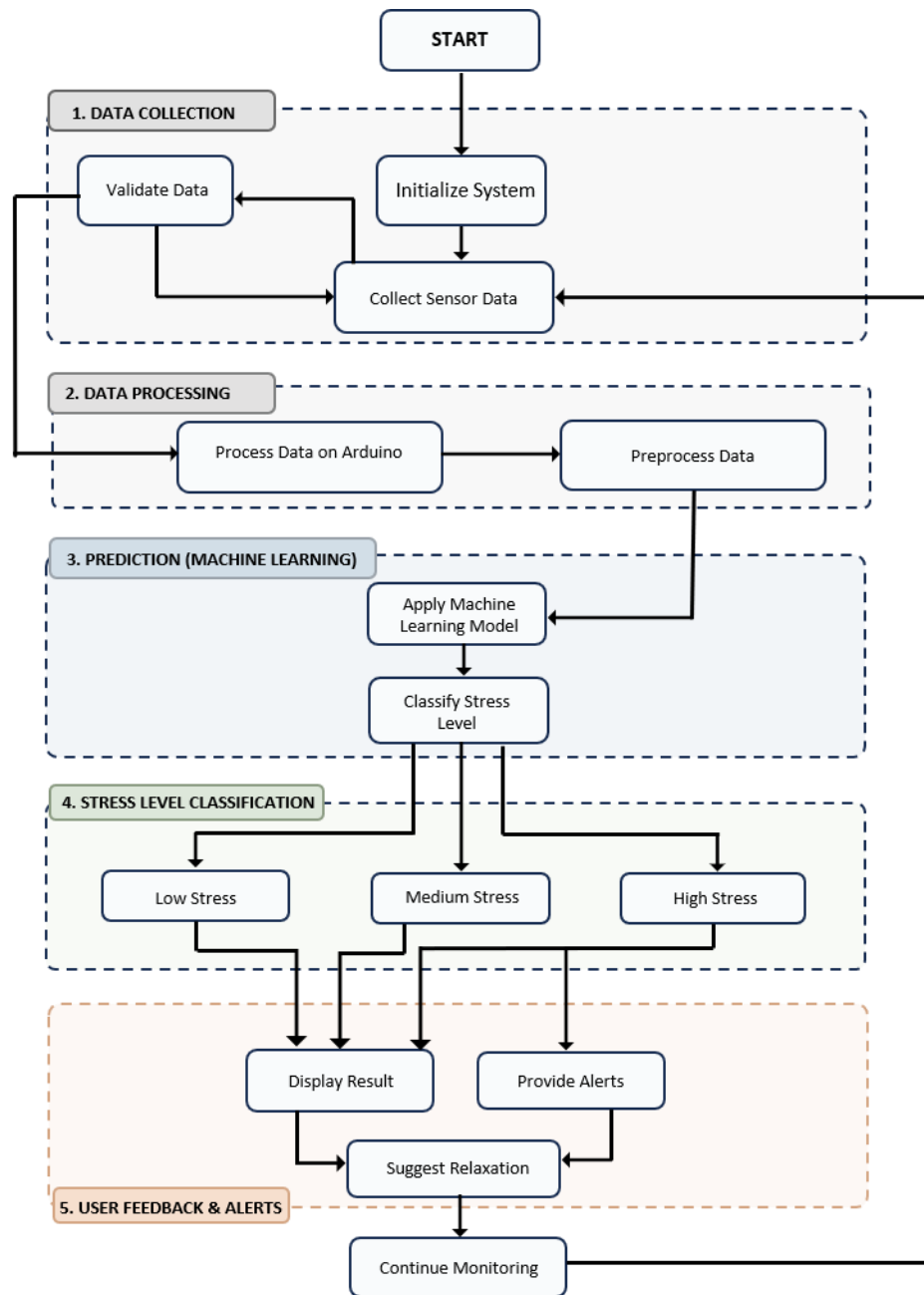
The Flask-based backend handled communication between the frontend dashboard, Firebase authentication module, and sensor-processing IoT components. Firebase integration enabled secure user authentication and real-time storage of monitoring records. Automated report generation functionality also allowed users to review previous stress observations and monitoring statistics.

The result suggests that combining multiple physiological and behavioral features improves the effectiveness of stress monitoring when compared to relying on a single parameter alone. The multi parameter design of Relaxo-Monitor provides improved accuracy by 82% and supports real-time monitoring.

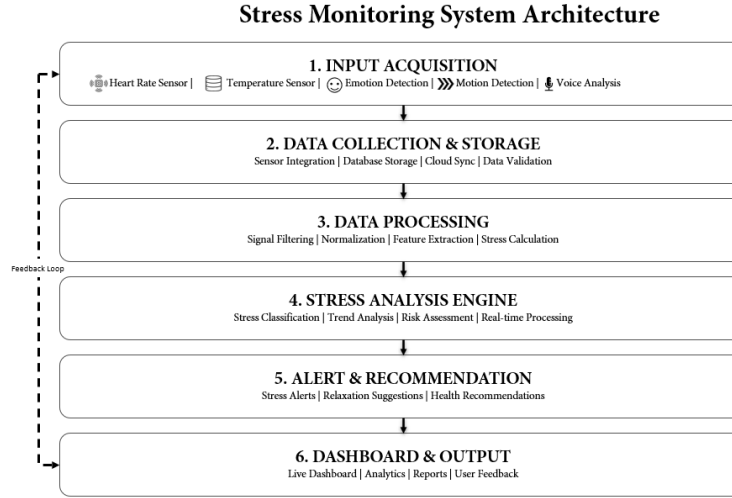
Although the system demonstrated real-time monitoring capabilities, several limitations remain. Sensor readings like occasionally fluctuate due to environmental conditions, improper sensor placement, or user movement. In addition, the current implementation relies primarily on a basic threshold-based Logistic regression. Class 2 also failed to predict by the model due to overlapping features. Future improvements may include the incorporation of more sophisticated machine learning models, larger datasets, cloud-based analytics, and improved wearable integration for enhanced stress prediction accuracy.

## 5 Figures

### 5.1 Figures



**Fig 1.** Flowchart representation of the Relaxo-Monitor system illustrating data collection, preprocessing, machine learning-based stress prediction, stress level classification, alert generation, and user feedback process



**Fig 2.** Overall architecture and workflow of the proposed Relaxo-Monitor system for real-time stress monitoring and analysis.

## 6 Statements

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**Availability of Data and Materials:** The data supporting the findings of this study are available within the article and its supplementary materials. Additional implementation details, dashboard outputs, and experimental observations may be obtained from the corresponding author upon reasonable request.

**Ethics Approval:** This study does not involve human participants, animal experimentation, or clinical trials. Therefore, ethical approval was not required for this work.

**Conflicts of Interest:** The authors declare no conflicts of interest related to this research work. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.



## 7 Abbreviations

The following abbreviations are used in this manuscript:

| Abbreviation | Full form                |
|--------------|--------------------------|
| AI           | Artificial Intelligence  |
| ML           | Machine Learning         |
| IoT          | Internet of Things       |
| BPM          | Beats Per Minute         |
| HRV          | Heart Rate Variation     |
| RMS          | Root Mean Square         |
| UI           | User Interface           |
| PDF          | Portable Document Format |
| SMS          | Short Message Services   |

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